Houston Karnen

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PROFESSIONAL SUMMARY

Graduate Master of IT (Distributed Computing) student passionate in backend and infrastructure. Two years of hands-on experience across full-stack web, distributed systems, cloud computing, and DevOps. Comfortable from design to deployment – whether it's a React front-end, Node back-end, or Terraform pipeline. Looking for a graduate software-engineering / infrastructure role where I can build scalable services and keep learning.

EDUCATION – University of Melbourne

Master of Information Technology - Distributed Computing (Distinction) *Jan 2024 – July 2025*
Key Areas: Distributed Algortihms & Systems, Cluster & Cloud Computing
Bachelor of Science - Computing & Software Systems *Jan 2021 – Dec 2023*
Key Areas: Algorithms, Databases, Software Design, Cybersecurity

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, SQL
Frameworks & Libraries: React, Tailwind, Node.js, Express, Flask, Spring Boot, XState, MPI
Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC), Azure (App Service, Storage), Kubernetes, Docker, Fission, OpenStack,
IaC & DevOps: Terraform, Git, GitHub Actions, SLURM, CI/CD, Agile
Data& ML: PostgreSQL, MySQL, MongoDB, Elasticsearch, LangChain, HuggingFace
Tools: Jira, Confluence, Postman, Jupyter, Unix / Linux CLI

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Cloud Practitioner *Ongoing*
- Microsoft Certified: Azure Fundamentals (AZ-900) *2025*
- Melbourne Employability Accelerator *2023*
- Melbourne International Undergraduate Scholarship *2021*

EXPERIENCE – Work Placements

Full Stack Developer *Mar 2025 – July 2025*
Clinical Reasoning Simulations – University of Melbourne x University of Eastern Finland

- Developed deterministic simulation engine using **YAML + XState v5**, modelling medical scenarios in real time and serving 50+ concurrent learners
- Integrated **LangChain & LLaMA 3** chatbot; 95% accuracy mapping multilingual text to actions
- Exposed RESTful APIs on **Node.js** secured with **JWT**, with 80% **Jest** coverage
- Added xAPI logging on **PostgreSQL**; wrote onboarding docs and Postman suites

Intern / Full Stack Developer *June 2024 – Dec 2024*
Queue Solutions

- Refactored legacy jQuery modules into **React + TypeScript**
- Built motivational-modelling graph tool with MaxGraph, used by UniMelb IT capstone students
- Practiced two-week Scrum sprints; led backlog refinement and demo sessions with client

Web Developer *July 2023 – Oct 2023*
Longbeach Community Centre

- Implemented **MERN**-based admin portal that digitized class attendance and tutor / admin controls; eliminated all paper records for 20+ weekly classes

Co-Founder / Full Stack Developer *Jan 2024 – Apr 2024*
StudyGenie

- Bootstrapped Azure-funded SaaS that parses course materials and auto-generates personalized quizzes via **Azure OpenAI**
- Deployed Flask REST API + React dashboard on **Azure App Services** and **Azure Storage**
- Integrated **Google OAuth2** for role-based access; onboarded 50+ pilot users

SELECTED PROJECTS

Backend Developer – To-Do List REST API (Personal) *Aug 2025 – Present*

- Built **Spring Boot** RESTful service with CRUD operations, pagination, and filtering
- Layered architecture with **Spring Data JPA** and **H2 / MySQL** persistence
- Validation and global exception handling; documented endpoints with **Swagger / OpenAPI**; added integration tests using **JUnit**

DevOps / Full Stack Developer – AWS Portfolio Deployment (Personal) *July 2025 – Aug 2025*

- Containerized React + TypeScript e-portfolio in **Docker**, deployed on AWS
- Provisioned VPC, public subnet, IGW, security group, EC2 (t3.micro) using **Terraform** IaC
- Automated first-boot environment setup (Docker install, repo clone) with cloud-init scripts

Full Stack Developer – Chrome Translation Extension (Personal) *July 2025 – Aug 2025*

- Developed Manifest V3 Chrome Extension that processes images of Korean novels directly
- Integrated pipeline: image capture -> **Tesseract.js** OCR -> **OpenAI GPT 4** translation, enabling instant text translation
- Designed user-friendly features for image selection, preview, and translation output
- Contributed documentation guidelines upstream to Tesseract.js repository

NLP Data Engineer – Fact-Checking Pipeline *Apr 2025 – June 2025*

- Two-stage retrieval system using Sentence-BERT + FAISS for semantic indexing of 1M+ documents with cross-encoder reranking and dynamic filtering boosting recall by 18%
- Used two-shot in-context learning and COT prompting with Gemma-3B LLM to improve reasoning and classification accuracy

DevOps / Cloud Engineer – Traffic & Air Quality Analytics *Apr 2024 – June 2024*

- Built serverless ETL pipeline on **Kubernetes + Fission (MRC)** to ingest hourly traffic (VicRoads) and air quality (BoM) datasets
- Designed data schemas for traffic sensor feeds and environmental measurements for consistent storage and querying in Elasticsearch
- Implemented data validation and cleaning during ingestion; correlations visualized on Jupyter

Data Engineer – SPARTAN HPC Cluster *Mar 2024 – Apr 2024*

- Parallel-processed 120GB JSON Twitter dataset with **mpi4py** + **ijson**; achieved 3x throughput with round-robin
- Automated benchmarking and job scheduling via SLURM batch scheduling across cluster configurations (1N1C, 1N8C, 2N8C)

Java Developer – Collaborative Whiteboard *Mar 2024 – May 2024*

- Designed and implemented a multithreaded server coordinating drawing and chat features for 10+ clients
- Added save/load, permission management, and network fault tolerance; scored 97%

Languages

- English (Fluent)
- Indonesian (Native)
- Chinese (Intermediate)

References available upon request